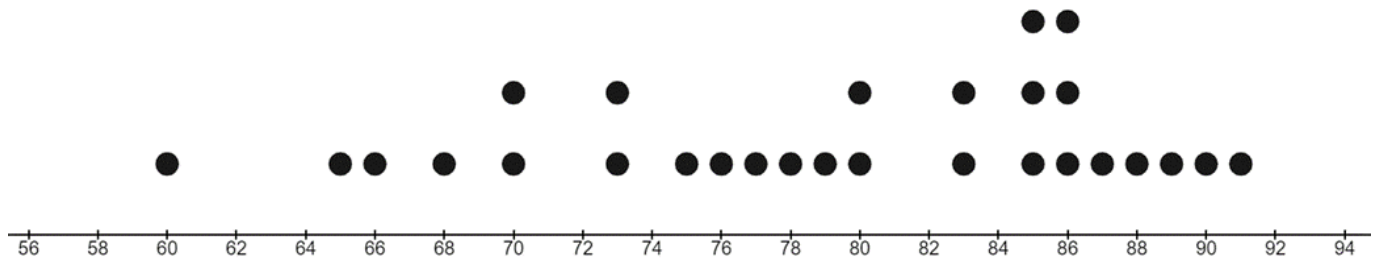


Grade 6
Unit 9
Lesson 9 Practice

Name _____
Date _____
Period _____

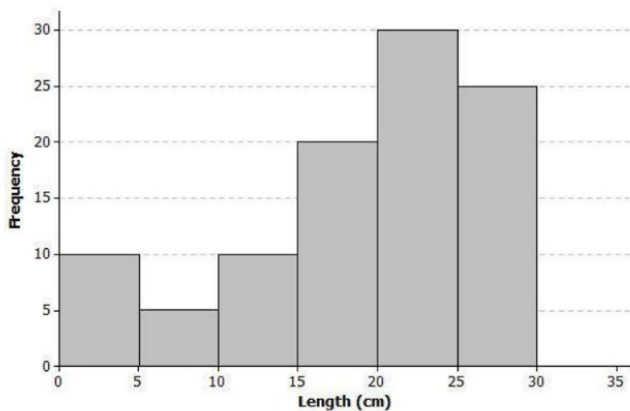
The grades (based on 100 points) of a class of students is shown in the dot plot.



- Describe the spread and shape of the distribution.
The distribution is spread out and slightly left skewed.
- Find and interpret the mean scores. Round to the nearest tenth.
$$[60 + 65 + 66 + 68 + 2(70) + 2(73) + 75 + 76 + 77 + 78 + 79 + 2(80) + 2(83) + 3(85) + 3(86) + 87 + 88 + 89 + 90 + 91] \div 28 = \frac{2214}{28} \approx 79.1$$

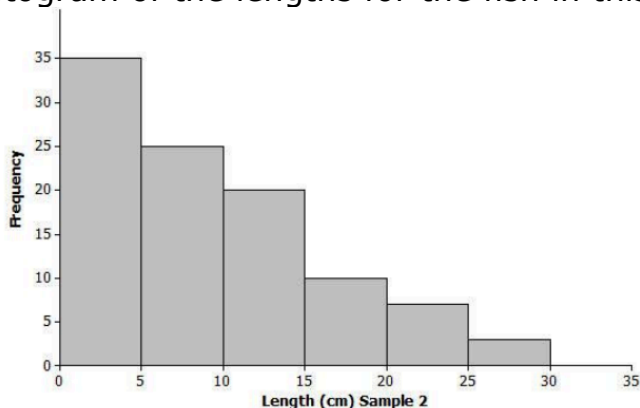
The average score of class A is 79.1.
- Find and interpret the median scores.
The median score was 80. Half of the scores were higher, and half of the scores were lower.
- How would the mean and median change if you found the mean and median of the top 10 grades on the test?
Both mean and median would increase.
- Find the mean of the top 10 scores.
$$\frac{2(85) + 3(86) + 87 + 88 + 89 + 90 + 91}{10} = \frac{873}{10} = 87.3$$
- Find the median of the top 10 scores.
86.5

Scientists collected data from many yellow perch because they were concerned about the survival of the yellow perch. Scientists captured yellow perch from a local lake. They recorded data on each fish and then returned each fish to the lake. Consider the following histogram of data on the length (in centimeters) for a sample.



7. Which measure of the center can be determined by the histogram, the mean or the median?
Neither can be exactly determined.
8. Estimate the median length of the yellow perch in the sample.
Answers will vary. Between 20 – 25 cm.
9. Based on the shape of this data distribution, do you think the mean length of a yellow perch would be greater than, less than, or the same as your estimate of the median?
Less than

Another sample of Great Lake yellow perch from a different lake was collected. A histogram of the lengths for the fish in this sample is shown below.



10. Based on the shape of this data distribution, do you think the median length of a yellow perch from this second sample would be greater than, less than, or the same the median length from the first sample?
The median of the second would be less than the first.